

NAME

saw – get the standard atomic weight

LIBRARY

ciaaw library (*libciaaw*, *-lciaaw*)

SYNOPSIS

```
Fortran      function saw(s,ab,u)result(res)  
              character(len=*), intent(in) :: s  
              logical, intent(in), optional :: ab  
              logical, intent(in), optional :: u  
              real(dp) :: res  
  
C            double ciaaw_saw(char *s,int n,bool ab,bool u);  
  
Python      saw(s:str,abridged:bool=True,uncertainty:bool=False)->float:
```

DESCRIPTION

Get the standard atomic weight for the element *s*. The default behavior is to provide the abridged value, *ab* is *TRUE*, recommended by the CIAAW. The uncertainty can be retrieved by setting *u*=*TRUE*.

Arguments:

<i>s</i>	Element symbol.
<i>ab</i>	Set to False if the abridged value is not desired. Default to <i>TRUE</i> .
<i>u</i>	Set to True if the uncertainty is desired. Default to <i>FALSE</i> .

RETURN VALUE

NaN	If the provided element is incorrect.
-1	If the element does not have a SAW.

EXAMPLES

```
Fortran:  
    use ciaaw  
    saw("H", ab=.false., u=.true.)
```

```
C:  
    include <stdio.h>  
    include "ciaaw.h"  
    ciaaw_saw("H", 1, false, true)
```

```
Python:  
    import pyciaaw
```

```
pyciaaw.saw("H", ab=False, u=True)
```

SEE ALSO

***ciaaw*(3), *ciaaw_cli*(1), *ciaaw_version*(3), *ciaaw_saw*(3), *ciaaw_ice*(3), *ciaaw_nice*(3), *ciaaw_naw*(3), *ciaaw_nnaw*(3), *ciaaw_ices*(3), *ciaaw_naws*(3), *ciaaw_s2z*(3)**